

Windows & Linux Command Line Cheat Sheet

A practical quick-reference for common commands used in basic administration, troubleshooting, and study review.

Tip: Windows commands below mix classic Command Prompt and PowerShell. Linux examples assume a bash-like shell.

Windows Command Prompt

Command	What it does	Example
ipconfig	Show IP configuration for adapters.	ipconfig /all
ping	Test basic network connectivity to another host.	ping 8.8.8.8
tracert	Trace the path packets take to a destination.	tracert google.com
nslookup	Query DNS records and name resolution.	nslookup openai.com
netstat -ano	List listening ports and active connections with PID.	netstat -ano
tasklist	Show running processes.	tasklist
taskkill /PID 1234 /F	Force-stop a process by PID.	taskkill /PID 1234 /F
dir	List files and folders in the current directory.	dir
cd	Change directories.	cd C:\Windows
mkdir	Create a new folder.	mkdir Logs
copy	Copy files.	copy report.txt backup.txt
del	Delete files.	del old.log

Windows PowerShell

Command	What it does	Example
Get-Help	Show help for PowerShell commands.	Get-Help Get-Process
Get-Process	List running processes.	Get-Process
Stop-Process -Id 1234	Stop a process by PID.	Stop-Process -Id 1234

Command	What it does	Example
Get-Service	List services and their status.	Get-Service
Restart-Service	Restart a Windows service.	Restart-Service Spooler
Test-Connection	Ping a host from PowerShell.	Test-Connection 8.8.8.8
Get-NetIPAddress	Show local IP addressing information.	Get-NetIPAddress
Get-Content	Read text from a file.	Get-Content .\notes.txt

Linux Shell Basics

Command	What it does	Example
pwd	Print current working directory.	pwd
ls -la	List files, including hidden ones and details.	ls -la
cd /path	Change directories.	cd /var/log
mkdir dir	Create a directory.	mkdir backups
cp src dst	Copy a file or directory.	cp file.txt backup.txt
mv src dst	Move or rename a file.	mv old.txt new.txt
rm file	Delete a file. Use carefully.	rm notes.txt
cat file	Print file contents.	cat /etc/hosts
less file	View a file page by page.	less /var/log/syslog
grep text file	Search for matching text.	grep error app.log
find /path -name	Search for files by name.	find /home -name report.txt
chmod	Change file permissions.	chmod 755 script.sh

Linux Networking & Process Commands

Command	What it does	Example
ip a	Show IP addresses and interfaces.	ip a
ss -tulpn	Show listening sockets and owning processes.	ss -tulpn
ping	Test network reachability.	ping 8.8.8.8

Command	What it does	Example
tracert	Trace packet path to a host.	tracert google.com
dig	Query DNS records.	dig openai.com
nslookup	DNS lookup utility.	nslookup openai.com
curl	Fetch data from a URL.	curl https://example.com
wget	Download a file from the web.	wget https://example.com/file.zip
ps aux	List running processes.	ps aux
kill -9 PID	Force-stop a process by PID.	kill -9 1234
top	Interactive process viewer.	top
sudo	Run a command with elevated privileges.	sudo systemctl restart ssh

Study hint: Focus first on what category a command belongs to - files, networking, DNS, processes, permissions, or services. That makes memorizing the exact syntax much easier.